🎓 **Exam Type: Practical**
⏳ **Duration: 6 Hours**

"Customer Churn Prediction — Multi-Algorithm Classification Showdown"

Project Overview

You are working as a Junior Data Scientist at a telecom company similar to Jio or Airtel. The retention team has escalated a critical business problem: the company is losing paying subscribers every month and cannot predict who is about to leave.

You are given the IBM Telco Customer Churn dataset — a widely used industry benchmark that mirrors real subscriber data. Your task is to build and compare multiple Supervised Learning classification models to predict whether a customer will churn (leave the service), and help the retention team decide which model to deploy in production.

📁 **Dataset: IBM Telco Customer Churn Dataset**

🔗 Kaggle Link: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

📊 Rows: 7,043 customers | Features: 21 columns (demographics, services, contract, billing, churn)

🎯 Target Column: Churn (Yes = churned, No = retained) — binary classification

⚠️ Note: This is a deliberately small, clean dataset — the challenge is in model selection, tuning, and interpretation, not data volume.

🎯 Learning Objectives

By completing this exam, students will demonstrate:

- Framing a real-world churn prediction problem from a business and ML perspective.
- EDA on a small, structured, real-world dataset with mixed feature types.
- Hands-on implementation of KNN, Naive Bayes, SVM, and Decision Tree classifiers.
- Handling class imbalance using class_weight and SMOTE.
- Rigorous model evaluation: confusion matrix, precision-recall tradeoff, ROC-AUC.
- Choosing the right model based on business cost of false negatives vs false positives.
- Packaging and submitting a clean reproducible ML project via GitHub.

🧠 Step 1: Problem Framing & Theory Notes

Write the following as Markdown cells in your notebook before coding:

1. What is Customer Churn? Why is predicting churn valuable for a business like Jio or Airtel? (Mention CAC vs CLV in your answer.)
2. Define the four cells of a Confusion Matrix (TP, TN, FP, FN). In the churn context — what is the business cost of a False Negative? What about a False Positive?
3. What is Class Imbalance? Does the Telco Churn dataset have it? How does SMOTE address it?
4. Briefly explain how each of the 4 algorithms you will use works — KNN, Naive Bayes, SVM, Decision Tree (2–3 lines each).
5. When would you choose Precision over Recall as your primary metric? Give a churn-specific answer.

> 💡 Tip: Keep theory cells concise — 2 to 4 bullet points per question. Real understanding counts more than long paragraphs.

📂 Step 2: Dataset Loading & Exploratory Data Analysis

2.1 Load & Inspect

- Load WA_Fn-UseC_-Telco-Customer-Churn.csv using pandas.
- Print .shape, .info(), .describe(), and .head(10).
- The TotalCharges column loads as object — convert it to float64. Investigate why (hint: whitespace rows) and handle them.
- Check and report class balance: What % of customers churned? Is this imbalanced?

2.2 Univariate Analysis

- Plot a countplot of the target variable Churn.
- Plot histograms for: tenure, MonthlyCharges, TotalCharges.
- Plot countplots for key categorical features: Contract, InternetService, PaymentMethod.

2.3 Bivariate Analysis

- Churn rate by Contract type — which contract type churns the most? Plot a grouped bar chart.
- Churn rate by tenure bucket (bin tenure into 0–12, 13–24, 25–48, 49–72 months). Plot a bar chart.
- Boxplot: MonthlyCharges vs Churn. Do churners pay more per month?
- Heatmap: Correlation matrix of all numerical features. Note any strong correlations.

> 📌 Key insight to look for: Month-to-month contract customers with high MonthlyCharges and low tenure are the highest-risk churn segment. Does your EDA confirm this?

🔧 Step 3: Data Preprocessing & Feature Engineering

3.1 Drop & Clean

- Drop the customerID column (unique identifier, no predictive value).
- Handle the TotalCharges nulls/whitespace identified in Step 2 — fill with median.

3.2 Feature Engineering

- Create tenure_group: bin tenure into 4 groups — 'New' (0–12), 'Mid' (13–36), 'Senior' (37–60), 'Loyal' (61+).
- Create num_services: count the number of add-on services each customer subscribes to (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies — count 'Yes' values).
- Create AutoPay: 1 if PaymentMethod contains 'automatic', else 0.

3.3 Encoding

- Binary columns (Partner, Dependents, PhoneService, PaperlessBilling, Churn, etc.) — map Yes→1, No→0.
- Apply One-Hot Encoding to: InternetService, Contract, PaymentMethod.
- Apply Label Encoding to: tenure_group (preserving ordinal order).
- Drop original columns that have been replaced by engineered features.

3.4 Scaling

- Apply StandardScaler to: tenure, MonthlyCharges, TotalCharges, num_services.
- Note: KNN and SVM are distance-based — scaling is critical. Decision Tree and Naive Bayes are not — but scale all features uniformly for a fair comparison.

3.5 Handle Class Imbalance

- Report class counts in the training set.
- Apply SMOTE (from imblearn) to the training set only. Print new class counts after oversampling.
- Important: Apply SMOTE only after the train-test split — never on the full dataset.

3.6 Train-Test Split

- Split: 80% train / 20% test, stratify=y, random_state=42.
- Print shapes of X_train, X_test, y_train, y_test.

> ⚠️ SMOTE Rule: Fit SMOTE only on X_train / y_train. The test set must remain original (unaugmented) to simulate real-world evaluation.

Step 4: Model Building — KNN & Naive Bayes

4.1 K-Nearest Neighbours (KNN)

- Train KNeighborsClassifier with k=5 as baseline. Report Accuracy, Precision, Recall, F1, AUC-ROC.
- Tune k: Run a loop for k = 1, 3, 5, 7, 9, 11, 15. Plot k vs F1-Score (on validation set). Which k is best?
- Re-train with the optimal k. Plot Confusion Matrix heatmap.
- Plot the ROC curve and report AUC.

4.2 Naive Bayes — Gaussian NB

- Train GaussianNB with default parameters. Report all metrics.
- Print the class_prior_ probabilities learned by the model. Do they match actual churn rate?
- Plot Confusion Matrix heatmap and ROC curve.
- Note: Naive Bayes assumes feature independence. Write 2 sentences on whether this assumption holds for this dataset.

> 💡 Reminder: For churn prediction, a False Negative (predicting 'No Churn' when customer actually leaves) is more costly than a False Positive. Focus your analysis on Recall for the Churn class.

Step 5: Model Building — SVM & Decision Tree

5.1 Support Vector Machine (SVM)

- Train SVC(kernel='rbf', C=1, gamma='scale', probability=True, random_state=42).
- Report: Accuracy, Precision, Recall, F1, AUC-ROC. Plot Confusion Matrix.
- Tune C: Try C = [0.1, 1, 10, 100] using 3-Fold cross_val_score (scoring='f1'). Plot C vs F1. Select best C.
- Re-train with best C. Report final test metrics and ROC curve.
- Note: SVM is a black-box model — write 2 sentences on the business tradeoff between interpretability and performance.

5.2 Decision Tree Classifier

- Train DecisionTreeClassifier(max_depth=5, class_weight='balanced', random_state=42).
- Report all metrics. Plot Confusion Matrix.
- Visualize the tree using plot_tree() (max depth shown = 3 for readability). What is the root split feature?
- Tune max_depth: Try depth = [3, 4, 5, 6, 7, 8, None] using cross_val_score (scoring='f1'). Plot depth vs F1.
- Re-train with best depth. Plot feature importances (top 15 features) as a sorted bar chart.

5.3 One Model with class_weight='balanced' vs SMOTE

- Take your best-performing model so far. Train it twice: once on SMOTE-augmented data, once on original data with class_weight='balanced'.
- Compare Recall for the Churn class between the two approaches. Which gives better Recall?

📈 Step 6: Model Evaluation & Comparison

Create a consolidated comparison table of all 4 models showing:

- Accuracy
- Precision — Churn class
- Recall — Churn class (most important!)
- F1-Score — Churn class
- AUC-ROC
- Training Time (seconds)

Present as a formatted pandas DataFrame sorted by Recall (descending).

- Plot all 4 ROC curves on a single figure with a legend.
- Plot a grouped bar chart comparing Precision vs Recall for all 4 models side by side.
- Write a recommendation (4–6 sentences): Which model would you deploy for Jio/Airtel's retention team, and why? Consider Recall, interpretability, and latency.

> 💡 Business framing: Jio's retention team can call ~500 customers/day. If the model flags 2,000 likely churners, they can only act on 25%. Precision matters too — but missing a real churner (False Negative) is the bigger loss.

🔍 Step 7: Error Analysis & Interpretation

7.1 Error Analysis on Best Model

- From your best model's predictions, isolate all False Negatives (actual churners predicted as No Churn).
- Profile the False Negatives: What is the average tenure, MonthlyCharges, and Contract type of these missed customers?
- Compare this profile to the overall churner profile. What pattern do you notice?

7.2 Feature Importance / Decision Insight

- For your Decision Tree: list the top 5 split features and explain what each means for churn prediction.
- Write 3–5 business-friendly bullet points. E.g.: 'Customers on Month-to-Month contracts appear in the first split — they are 3x more likely to churn than those on 2-year contracts.'
- (Optional Bonus) If you used SVM as your best model: compute permutation_importance from sklearn.inspection and plot top 10 features.

🚀 Step 8: Pipeline, Deployment & GitHub Submission

8.1 Save the Final Pipeline

- Wrap best model + preprocessor into a sklearn Pipeline.
- Save: joblib.dump(pipeline, 'churn_model.pkl').

- Test: load the pickle and predict churn probability for 5 sample customers. Print predicted probabilities and final label.

8.2 GitHub Submission

- Create a GitHub repository named: telco-churn-supervised-learning.
- Push the following:
  - CustomerChurn_SupervisedLearning.ipynb (fully executed notebook)
  - churn_model.pkl (saved pipeline)
  - summary_report.md (see below)
  - requirements.txt
  - README.md (project overview, dataset link, how to run)

8.3 Summary Report (summary_report.md)

Write ~400–500 words covering:

- Business problem and dataset summary.
- Preprocessing and class imbalance strategy used.
- Which model you recommend and why (with metric evidence).
- Top 3 churn signals found — what action should the retention team take?
- What would you do next: more data, deep learning, real-time scoring API?

Marking Scheme (50 Marks)

Total Marks: 50 — distributed across 3 components as follows:

Component	Marks	Deliverable
A. Technical Completion	30 Marks	Completed .ipynb & .pkl file with all tasks built and functional
B. Recorded Video (Face + Screen)	15 Marks	5-10 min MP4/MOV showing face + screen with verbal explanation
C. GitHub Repository + README.md	5 Marks	Public GitHub repo with screenshots, README, video link, commits
TOTAL	50 Marks	

Component B: Recorded Video - Face + Screen (15 Marks)

Record a video showing your face (webcam picture-in-picture) and your computer's entire screen simultaneously. Explain concepts verbally throughout — do not just click without speaking.

Video Requirements:

Requirement	Detail
Format	MP4 or MOV, maximum 500 MB

Face visibility	Webcam picture-in-picture overlay visible throughout the full recording
Screen content	Entire screen (by demonstrating ipynb and pkl file)
Duration	5 to 10 minutes
Naming convention	Practical_Exam_YourName_GRID.mp4
Upload location	Google Drive (Anyone with link can view) or YouTube (Unlisted)
Link placement	Paste the video URL in your GitHub README.md under the Video section

Component C: GitHub Repository + README.md (5 Marks)

Create a public GitHub repository specifically for this project. Add the following files as mentioned in the Deliverables section below.

Deliverables

Notebook	CustomerChurn_SupervisedLearning.ipynb (fully executed, all outputs visible)
Saved Model	churn_model.pkl (sklearn Pipeline)
Summary Report	summary_report.md (theory + observations, ~1 page)
GitHub Repo	All of the above + requirements.txt + README.md

✅ **Submission Checklist:**

- ☐ Notebook runs top-to-bottom without errors
- ☐ SMOTE applied only on training data — not full dataset
- ☐ Model comparison table present and sorted by Recall (Step 6)
- ☐ All 4 ROC curves plotted on one figure
- ☐ GitHub repo link shared with examiner
- ☐ README.md explains how to run the project

BRING ON YOUR CODING ATTITUDE

Practical Exam — Supervised Learning